

Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control

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Single Intersection control

Queue length as reward for single intersection [arXiv:1905.04722]
Learning phase competition [CIKM'19a]
Learning from demonstration [CIKM'19c]
Learning from other signals [AAAI'20b]

Multi-intersection control

Learning with graph attention [CIKM'19b]
Pressure as reward for arterial [KDD'19]
Pressure as reward for city [AAAI'20a]

Simulator

Efficient traffic simulator CityFlow [WWW'19]
Learning to simulate car following model [ICDE'20]

Introduction



Figure 1. Traffic Congestion in Manhattan

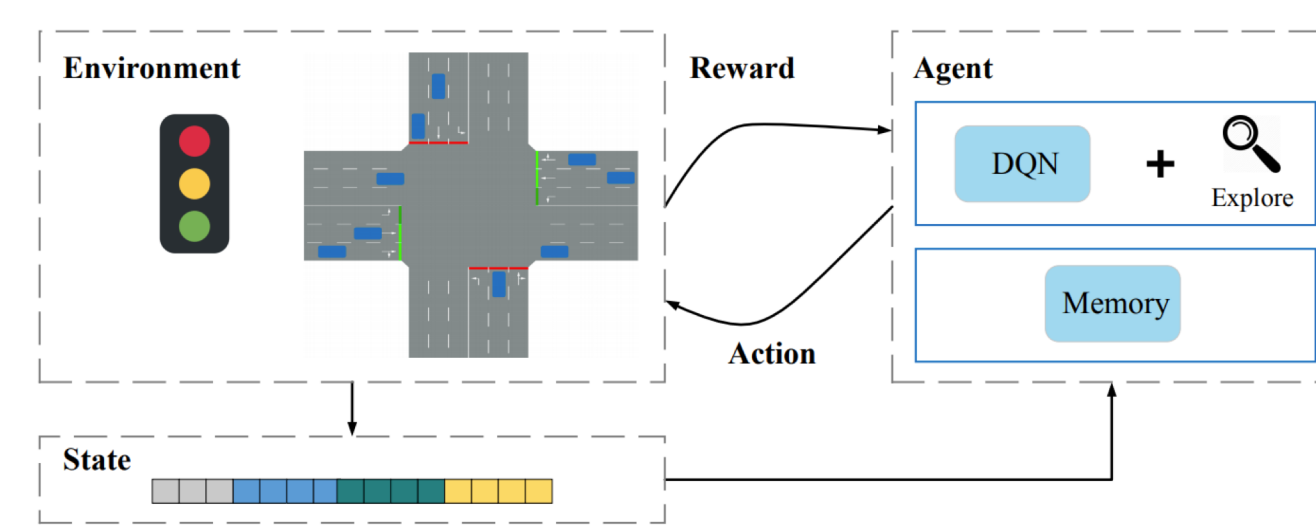
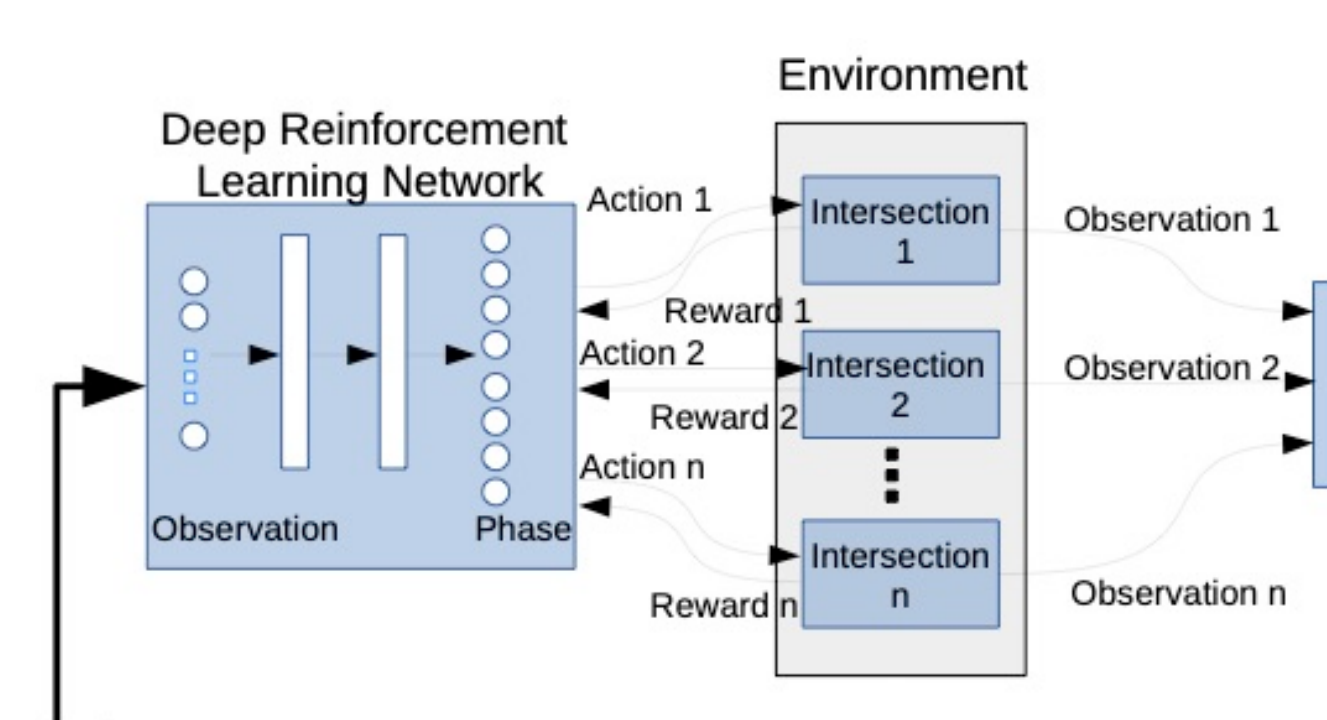


Figure 2. Reinforcement Traffic Signal Control Framework

Model framework



Key Features

- Scalability
- Signal coordination
- Data feasibility
- etc.

Key challenges

1. Scalability:

How to scale up to thousands of intersections?

2. Signal coordination:

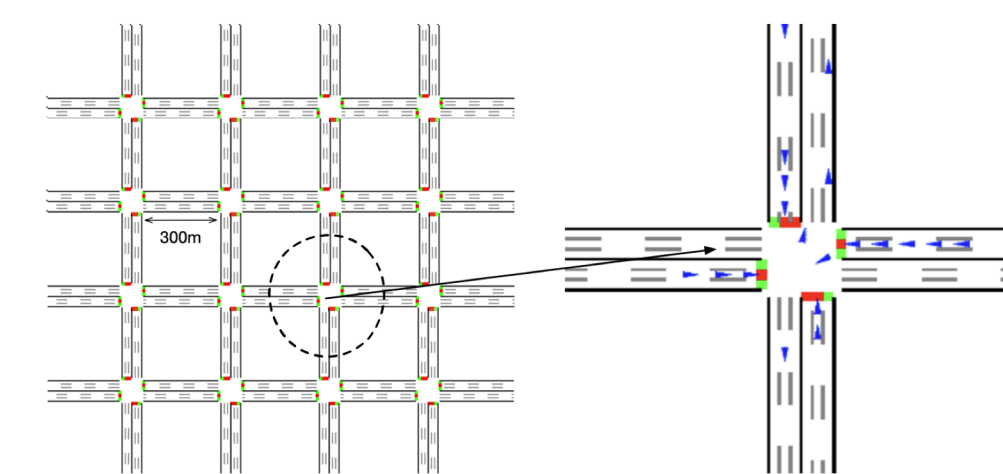
How to coordinate between intersections?

3. Data feasibility:

How to design the model from feasible data source

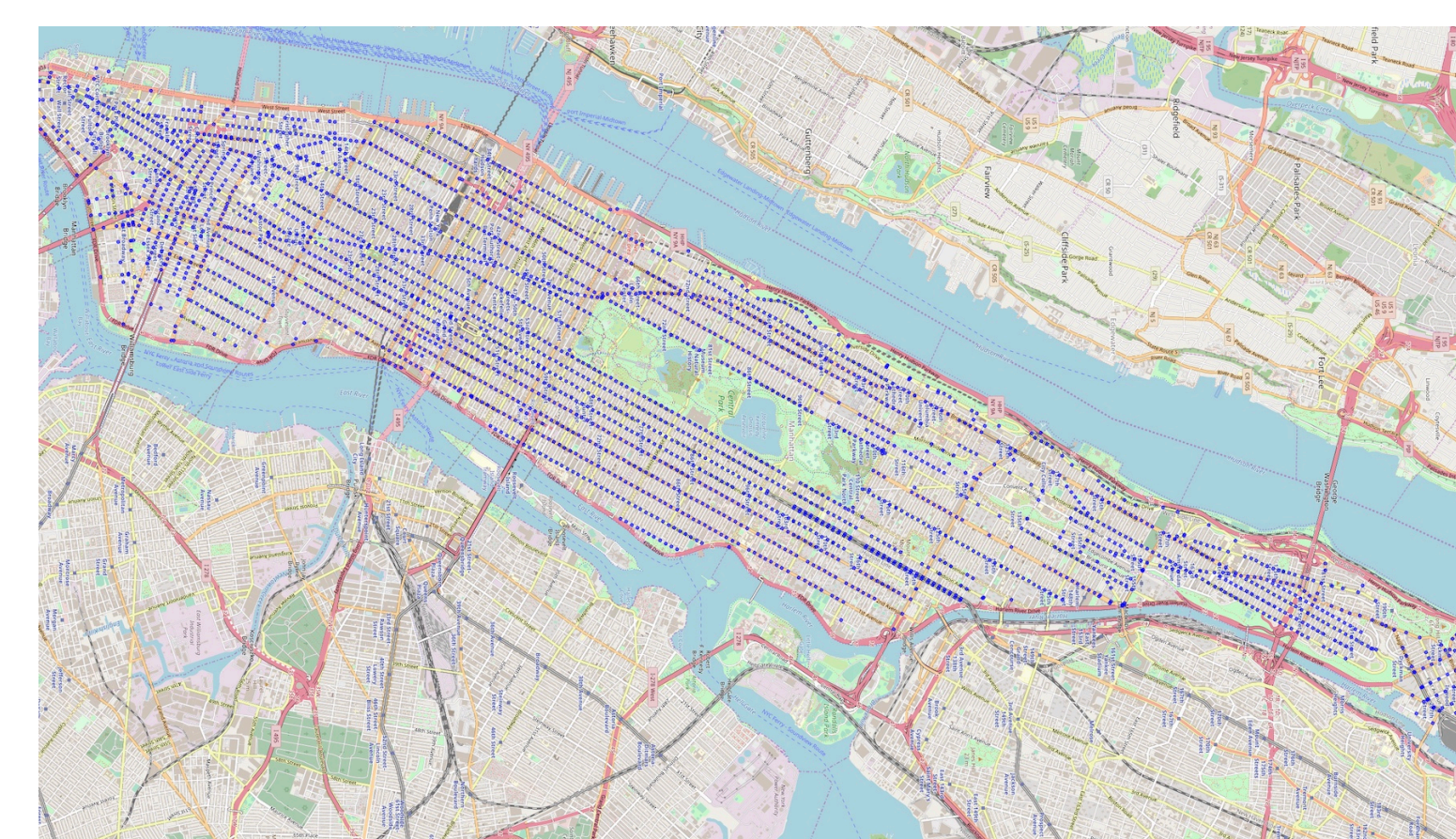
Experiments

4x4 roadnet



Model	Travel Time			
	Config 1	Config 2	Config 3	Config 4
FixedTime	573.13	564.02	536.04	563.06
MaxPressure	361.17	402.72	360.05	406.45
GRL	735.38	758.58	771.05	721.37
GCN	516.65	523.79	646.24	585.91
NeighborRL	690.87	687.27	781.24	791.44
PressLight	354.94	353.46	348.21	398.85
FRAP	340.44	298.55	361.36	598.52
MPLight	309.33	262.50	281.34	353.13

Manhattan roadnet



Model	Travel Time
FixedTime	974.23
MaxPressure	497.76
GRL	-*
GCN	653.45
NeighborRL	-*
PressLight	600.42
FRAP	512.70
MPLight	472.51

Methodology

Challenge #1: Frap base model & parameter sharing

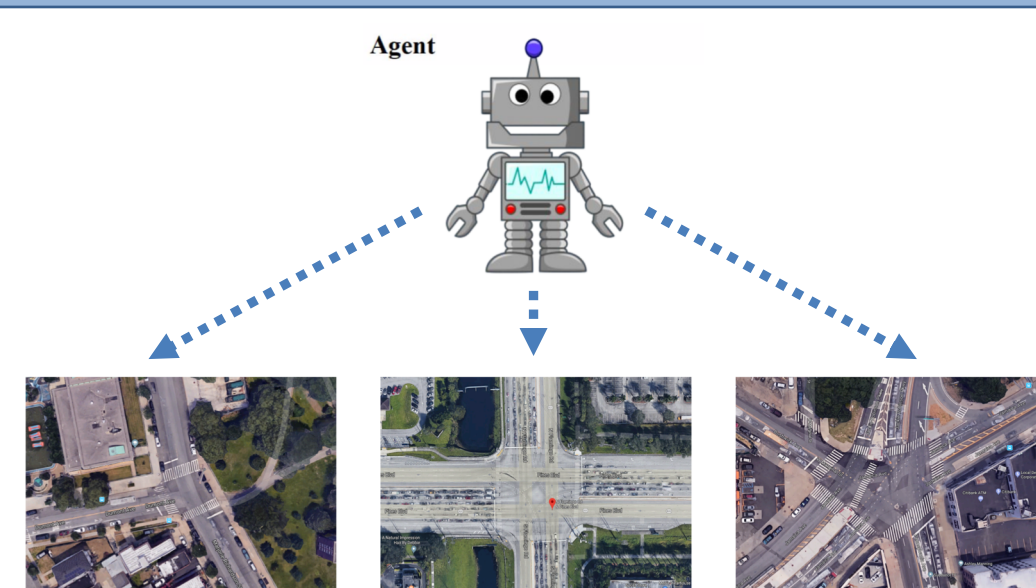


Figure 3. Frap base model for heterogeneous intersections [CIKM'19a]

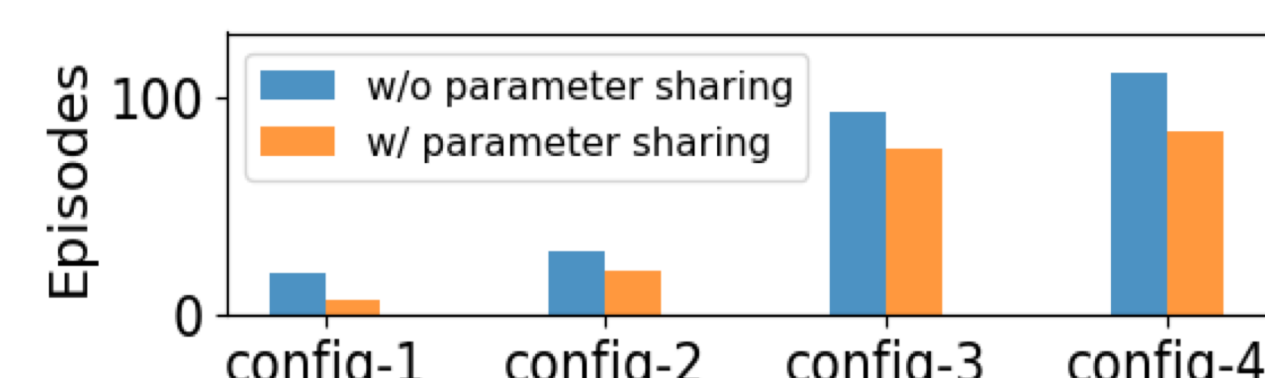
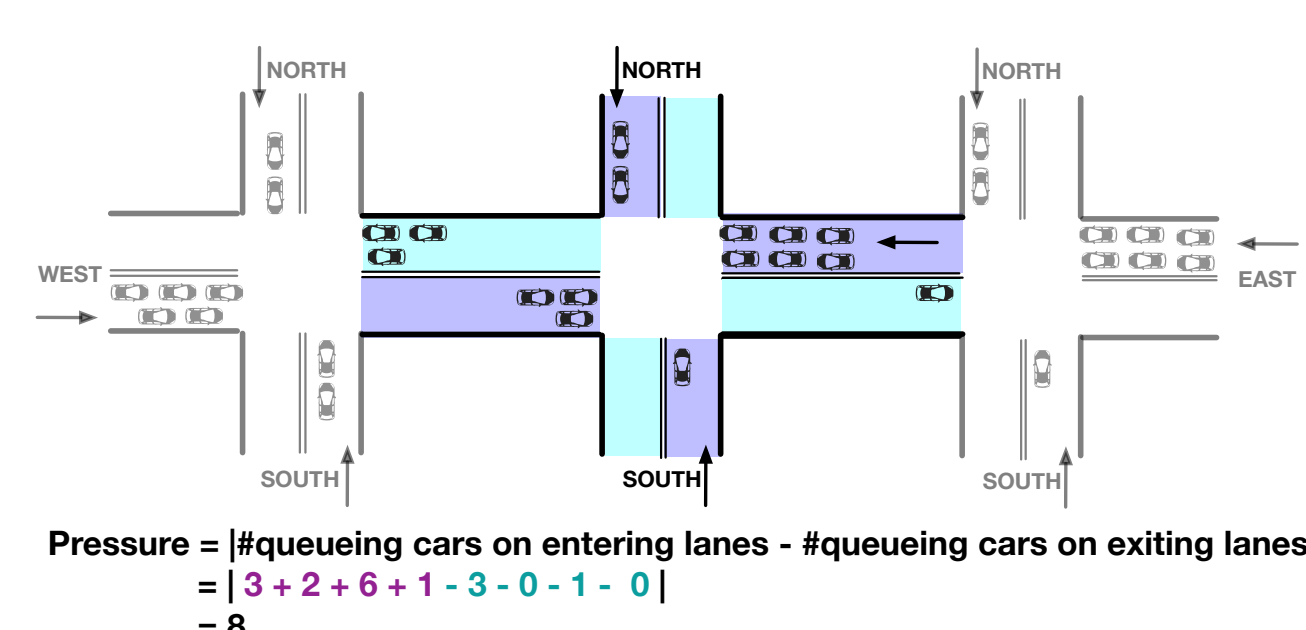


Figure 4. Leveraging parameter sharing

Challenge #2: Pressure-based coordination



$$\text{Pressure} = \# \text{queueing cars on entering lanes} - \# \text{queueing cars on exiting lanes}$$

$$= |3 + 2 + 6 + 1 - 3 - 0 - 1 - 0|$$

$$= 8$$

Challenge #3: simple state and reward

State:

- Current phase
- # Vehicles on incoming lanes
- # Vehicles on outgoing lanes

Reward:

Pressure for each intersection

Acknowledgements

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References

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[WWW'19] Zhang et al., CityFlow: A Multi-Agent Reinforcement Learning Environment for Large Scale City Traffic Scenario



Project Website

Find datasets, code, demo and more!